

Problem

Let a, b, c, d, e be positive integers satisfying

$$abcde = a + b + c + d + e.$$

Find the maximum possible value of

$$\max\{a, b, c, d, e\}.$$

Source: Titu Andreescu

§1 Bounding the Product of the Smaller Variables

Without loss of generality, arrange the variables so that

$$a \leq b \leq c \leq d \leq e.$$

We seek the largest possible value of e .

Since all variables are positive,

$$e < a + b + c + d + e = abcde.$$

On the other hand,

$$a + b + c + d + e \leq 5e,$$

and therefore

$$abcde < 5e.$$

Combining these inequalities gives

$$e < abcde < 5e.$$

Since $e > 0$, dividing through by e yields

$$1 < abcd < 5.$$

Because a, b, c, d are positive integers, the only possible values of their product are

$$abcd \in \{2, 3, 4\}.$$

Consequently, the only possibilities for (a, b, c, d) are

$$(1, 1, 1, 2), \quad (1, 1, 1, 3), \quad (1, 1, 1, 4), \quad (1, 1, 2, 2).$$

§2 Determining the Corresponding Values of e

Substituting each possibility into

$$abcde = a + b + c + d + e$$

gives:

$$2e = 5 + e \implies e = 5,$$

$$3e = 6 + e \implies e = 3,$$

$$4e = 7 + e \implies e = \frac{7}{3},$$

which is not an integer,

and

$$4e = 6 + e \implies e = 2.$$

Thus the largest attainable value is

$$e = 5.$$

§3 Verification of the Extremal Case

The tuple

$$(a, b, c, d, e) = (1, 1, 1, 2, 5)$$

indeed satisfies

$$abcde = 1 \cdot 1 \cdot 1 \cdot 2 \cdot 5 = 10$$

and

$$a + b + c + d + e = 1 + 1 + 1 + 2 + 5 = 10.$$

Hence it is a valid solution.

§4 The Final Conclusion

Therefore,

$$\boxed{\max\{a, b, c, d, e\} = 5}.$$